KENBRIGHT ACTUARIAL AND FINANCIAL SERVICES LIMITED

DEFINED BENEFITS VALUATION – DATA CHECKS

INTERNAL TECHNICAL PROCEDURES

VERSION 1.0

JANUARY 2026

KAFS POLICY

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OUR REF

\\Kenbright\Internal Technical Procedures\Defined Benefits Valuation Manual

DOCUMENT PURPOSE

This document has been produced as part of the internal technical procedures for KAFS and provides guidance on valuation processes for Defined Benefit Calculations

DOCUMENT OWNERSHIP

The owner of this document is Kenbright Actuarial and Financial Services Limited

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ABBREVIATIONS

NAME	DEFINITION
AVC	Additional Voluntary Contributions
DB	Defined Benefit
DC	Define Contribution
ERA	Early Retirement Age
IPP	Individual Pension Plan
KAFS	Kenbright Actuarial and Financial Services Limited
LRA	Late Retirement Age
NRA	Normal Retirement Age
RBA	Retirement Benefits Authority
SRBS	Staff Retirement Benefits Scheme
TDR	Trust Deed & Rules

1. CLASSIFICATION OF SCHEMES

1.1. Pension schemes are typically employer-sponsored plans established to provide employees with income upon retirement. Retirement Benefits Schemes in Kenya can be classified by five distinct categories: Funding Type, Funding Structure, Product Type, Investment Type and Benefit Type

1.2. The table below shows a brief summary of the key features of the above schemes.

Category	Type	Description
Funding Type	**Contributory Schemes**	Both employer and employee make regular contributions, usually a % of salary. Contributions are invested to grow over time.
	Non-Contributory Schemes	Employer is solely responsible for funding retirement benefits. Employees have no obligation to contribute, but may do so voluntarily.
Funding Structure	**Defined Benefit (DB)**	Guarantees a specific retirement benefit, based on a formula (e.g., salary × service). Employer bears investment and funding risk. Requires periodic actuarial valuations. DB schemes are declining due to funding challenges.
	Defined Contribution (DC)	Employer and employee contribute fixed amounts (flat or % of salary). Benefits depend on accumulated contributions plus investment income, net of expenses. Member bears investment and longevity risk.
Product Type	**Occupational Schemes**	Stand-alone schemes established by an employer or group of related companies. Trustees (corporate or individual) manage the scheme. Can be DB or DC.
	Umbrella Schemes	Sponsored by an external provider for multiple employers. Benefits include shared admin costs, economies of scale, wider investment options. Limitation: employers have less control over investments.
	Individual Retirement Schemes	Established by a sponsor for individuals to save independently for retirement. Flexible, with tax advantages and investment benefits similar to employer-based schemes. Not tied to employment.

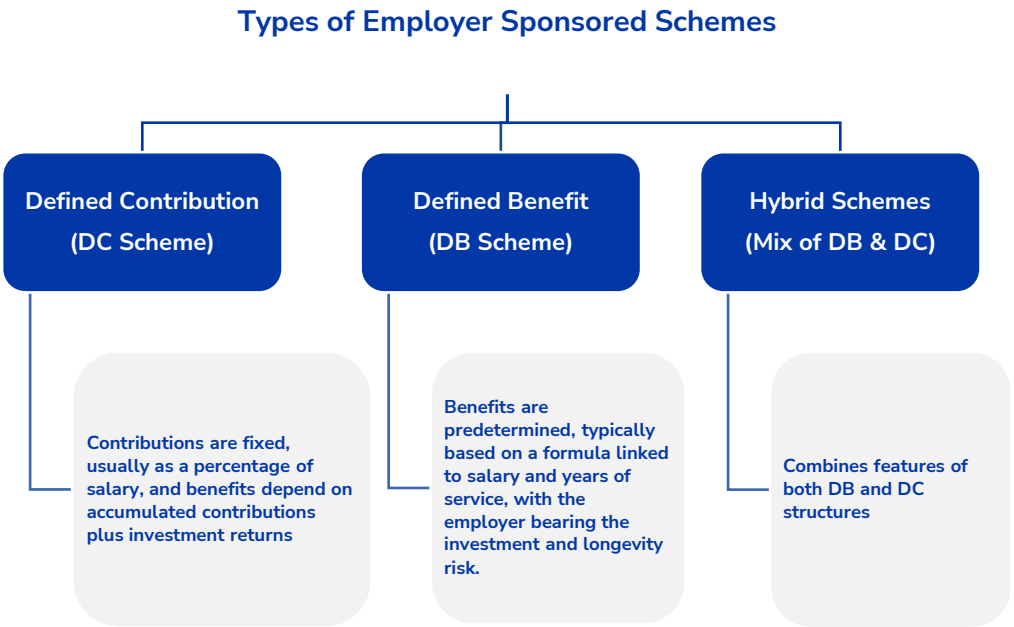
Category	Type	Description
Investment Type	**Guaranteed Schemes**	Capital and a minimum return are guaranteed. Managed by Approved Issuers (e.g., insurers). Conservative to moderate risk profile. Trustees receive a Deposit Administration Policy.
	Segregated Schemes	Investments are tailored to scheme's membership profile. No guarantee on capital or returns. Trustees and fund managers develop an investment strategy in line with the Investment Policy Statement.
	Hybrid Schemes	Combination of guaranteed and segregated funds. Members' balances are split between secure (guaranteed) and growth-oriented (segregated) investments. Flexibility may eventually extend to member-level choices.
Benefit Type	**Provident Fund**	Provides a lump sum at retirement. Member may choose to use part of the benefit to buy an annuity or income drawdown product.
	Pension Fund	Provides both a pension and partial lump sum. Up to one-third may be taken as cash, while the remaining two-thirds is used to purchase an annuity or fund an income drawdown.

2. TYPES OF SCHEMES

Overview

2.1. Pension schemes are typically employer-sponsored plans established to provide employees with income upon retirement.

2.2. These plans generally fall into three broad categories:



Defined Contribution Scheme

Overview

2.3. A Defined Contribution (DC) scheme is a type of pension arrangement where the **contributions are fixed**, usually as a percentage of the employee's salary, while the **benefits payable at retirement are not predetermined**.

2.4. Instead, retirement benefits depend on the total contributions made and the investment returns earned on those contributions.

2.5. In DC schemes, both the employer and employee typically make regular contributions into an individual account maintained for each member. The account balance grows through ongoing contributions and investment income, forming the basis of the member's retirement benefits.

2.6. In a Defined Contribution (DC) scheme, a member's retirement benefit is determined by the balance in their individual account. This balance reflects the total employer and employee contributions, less expenses, together with investment returns earned.

Key Features

2.7. The key features of a DC Scheme include:

a) **Contributions:** Fixed contributions by employer, employee, or both, commonly expressed as a percentage of pensionable salary.

b) **Individual Accounts:** Each member has a separate account where contributions and investment returns are accumulated.

c) **Benefit Uncertainty:** The final retirement benefit depends on factors such as the level of contributions, investment performance, and inflation over time.

d) **Risk Allocation:** The member bears both the investment risk (poor returns reduce benefits) and longevity risk (risk of outliving accumulated savings).

e) **Portability:** Balances are easily portable if a member changes employment, since funds are held in individual accounts.

Advantages of DC Scheme

2.8. The advantages of a DC Scheme include:

a) Transparent and straightforward to administer.

b) Flexible, as members can transfer balances between schemes when changing jobs.

c) Encourages savings discipline by building up benefits gradually.

Disadvantages of a DC Scheme

2.9. Disadvantages of a DC Scheme include:

a) Retirement income is uncertain and may be inadequate if contribution levels are low or if investments perform poorly.

b) Members need financial literacy to make informed decisions about contribution levels and investment choices.

c) Inflation and longevity risks may erode retirement income.

Key Distinction: Portability in DC vs. DB Schemes

- **Defined Contribution (DC) schemes**: Highly **portable**. Members can transfer their accumulated account balances to another registered scheme or to an individual retirement benefits (IPP's) scheme when they change jobs.
- **Defined Benefit (DB) schemes**: Benefits are generally **less portable**. Members who leave employment before retirement may preserve their accrued benefits in the scheme or transfer the *present value* of accrued rights, subject to scheme rules and regulatory approval.

Defined Benefit Scheme

Overview

2.10. A defined benefit (DB) Scheme is an arrangement where the benefits, which is ordinarily determined by the scheme rules, are defined in advance. Benefits are often related to the **final salary** and/or **years of service** of the employee.

2.11. Unlike Defined Contribution (DC) schemes, where benefits depend on investment performance, DB schemes promise members a predictable income at retirement.

2.12. The employer (or sponsor) is responsible for ensuring there are enough assets in the scheme to meet these promised benefits. As a result, the employer bears the **investment risk** (poor returns) and the **longevity risk** (members living longer than expected).

Key Features of DB Scheme

2.13. Key features of a DB Scheme include:

a) **Benefit Formula:** Commonly expressed as a percentage of final or average salary multiplied by years of service.

b) **Employer Responsibility:** Employer contributions are variable, determined by actuarial valuations to ensure benefits can be paid.

c) **Member Contributions:** May or may not be required, depending on scheme rules, but are usually fixed.

d) **Actuarial Valuation:** Required regularly to compare scheme assets against liabilities and determine funding status (surplus or deficit).

e) **Risk Allocation:** The employer bears the main financial risks, while members enjoy certainty of benefits.

Advantages of DB Scheme

2.14. The advantages of a DB Scheme include:

a) Predictable retirement benefits provide members with financial security.

b) Benefits often include ancillary provisions such as survivor pensions, death-in-service, or disability pensions.

c) Aligns retirement income more closely with salary progression and service.

Disadvantages of a DB Scheme

2.15. Disadvantages of a DB Scheme include:

a) Costly for employers, especially when investment performance is weak or members live longer than expected.

b) Requires complex actuarial valuations and regulatory oversight.

c) Benefits are less portable compared to DC schemes.

2.16. The Retirement Benefits Authority (RBA) mandates actuarial valuations at least once every **three years** for DB schemes to monitor funding levels and protect members' benefits.

Key Distinction: DC vs. DB Surpluses

- **Defined Contribution (DC) schemes**: Members do **not** have surpluses or deficits. Benefits equal the exact value of the individual account (contributions ± investment returns – expenses).
- **Defined Benefit (DB) schemes**: Surpluses or deficits arise because promised benefits are compared against scheme assets.

Hybrid Schemes

Overview

2.17. Hybrid pension schemes combine features of both Defined Contribution (DC) and Defined Benefit (DB) arrangements.

2.18. They are designed to balance the predictability of DB schemes with the flexibility and portability of DC schemes. In a hybrid structure, members may receive a minimum guaranteed benefit (DB element) while also accumulating an individual account that grows with contributions and investment returns (DC element).

2.19. Hybrid schemes are often used by employers who want to limit their long-term financial exposure but still provide employees with some level of benefit certainty.

Key Features

2.20. The key features of a Hybrid scheme include:

a) **Mixed Benefit Design:** May provide a guaranteed pension formula (DB) alongside an individual account (DC).

b) **Risk Sharing:** Investment and longevity risks are shared between the employer and members.

c) **Flexibility:** Members may access portable DC balances, while DB guarantees income security at retirement.

d) **Funding:** Requires careful actuarial valuation of the DB portion while maintaining record-keeping for the DC portion.

Types of Hybrid Designs

2.21. **Minimum Guarantee Plans** – The scheme guarantees a minimum DB pension, but members can receive more if DC investments perform strongly.

2.22. **Cash Balance Plans** – Employer promises a notional account balance that grows with contributions plus guaranteed interest credits.

2.23. **Combination Plans** – Members participate in both a DB formula and a separate DC account.

Advantages of Hybrid Schemes

2.24. Advantages of Hybrid Schemes include:

a) Provides employees with a degree of benefit certainty while still linking outcomes to investment performance.

b) Reduces the employer's long-term funding burden compared to pure DB schemes.

c) Can be structured to suit workforce needs and attract/retain talent.

Disadvantages of Hybrid Schemes

2.25. Disadvantages of Hybrid Schemes include:

a) More complex to design and administer than pure DB or DC schemes.

b) Requires both actuarial valuations (for DB portion) and account administration (for DC portion).

c) Communication to members must be clear to avoid confusion about how benefits accrue.

3. DATA REQUIREMENTS

Overview

3.1. Accurate and comprehensive data is fundamental to conducting an accurate valuation.

3.2. The quality of the data directly impacts the precision of liability calculations, the assessment of funding status, and compliance with regulatory requirements.

3.3. Incomplete or inconsistent data can lead to misstatements of scheme obligations, delays in reporting, and challenges in decision-making for trustees and regulators.

3.4. For the DB valuation as at the specified date, the policy level data required can be categorized into four key registers:

a) Membership Register (Active in-service members)

b) Deferred register (Deferred Members)

c) Dormant Members

d) Pensioners including dependants and principal members

3.5. For DC Valuation Withdrawals register would also be requested from the scheme.

Membership Register (Active in Service)

3.6. This contains a detailed record of all **active members** of the pension scheme. It provides essential information for benefit calculations, compliance, and reporting.

3.7. **Key Data Items:**

i. Member Number / Unique Identifier

ii. Name

iii. Gender

iv. Membership Status

v. Date of Birth

vi. Date of Employment

vii. Date of Joining Scheme

viii. Date of Exit (if applicable)

ix. Reason for Exit

x. Final Average Salary

xi. Accumulated Member Balance (as at valuation date)

Withdrawals Register

3.8. Tracks withdrawals or distributions made from participants' pension accounts.

3.9. This is particularly relevant for Defined Contribution (DC) schemes where members access individual accounts upon retirement, resignation, or other qualifying events.

3.10. **Key Data Items:**

i. Member Number / Unique Identifier
ii. Name
iii. Gender
iv. Membership Status
v. Date of Birth
vi. Date of Employment
vii. Date of Joining Scheme
viii. Date of Exit
ix. Reason for Exit
x. Withdrawal Amount (Gross and Net)
xi. Date of Withdrawal

Deferred Register

3.11. Maintains details of members in Defined Benefit (DB) schemes who have accrued pension rights but are yet to reach retirement age. This ensures accurate tracking of future benefit entitlements.

3.12. **Key Data Items:**

i. Member Number / Unique Identifier
ii. Name
iii. Gender
iv. Membership Status
v. Date of Birth
vi. Date of Employment
vii. Date of Joining Scheme
viii. Date of Deferral
ix. Annual Salary at Deferral
x. Deferred Amount

4. DATA REQUEST TEMPLATE

DEFINED BENEFIT VALUATION

SUMMARY OF REQUIRED DATA

General Information

Client:	XXX Staff Retirement Benefits Scheme
Project:	Actuarial Valuation - Defined Benefit Scheme
Analysis Date:	31st December 2024 (Example)

Requested Data

1	*Membership register as at the Valuation Date
2	*Withdrawals Register from last valuation to the Analysis Date
3	*Membership Schedule for pensions in payment (Main pensioners/Dependents)
4	Scheme Trust Deed and Rules including any amendments
5	Annual Financial Statements for the last three years
6	Annual Fund Manager/Investment Reports for the last three years
7	Management Accounts (including Asset details) as at 31 December 2024
8	Current Investment Policy Statement
9	Copies of the last two (2) Actuarial valuation reports
10	Membership data as used for the previous valuation
11	Outstanding benefit payments as at 31 December 2024 (Current Valuation Date)

**Refer to the "Membership Register" and "Withdrawals Register" tabs for detailed template*

Active Employee Data

Required Data Fields	
	Employee ID
	Employee Category
	Gender
	Date of Birth (dd-mm-yyyy)
	Date of Employment (dd-mm-yyyy)
	Date of Joining Scheme (dd-mm-yyyy)
	Monthly Pensionable Salary (as at scheme closure date)
	Monthly Pensionable Salary (as at 31-Dec-2024)
	Accumulated Contributions with interest (as at 31 December 2024)

Deferred Member Data

Required Data Fields	
	Member ID
To be provided for each year for FY 2022, FY 2023 & FY 2024 (Example- To be provided for the previous 3 years)	Category
	Gender
	Date of Birth (dd-mm-yyyy)
	Date of Joining Scheme (dd-mm-yyyy)
	Date of Exit (dd-mm-yyyy)
	Monthly Pensionable Salary (as at scheme closure date)
	Monthly Pensionable Salary (as at Deferral Date)
	Deferred Lumpsum Amount
	Deferred Pension Amount

Main Pensioners

Required Data Fields	
	Member ID
To be provided for each year for FY 2022, FY 2023 & FY 2024. (Example- To be provided for the previous 3 years)	Category
	Gender
	Date of Birth (dd-mm-yyyy)
	Date of Joining Scheme (dd-mm-yyyy)

	Date of Retirement (dd-mm-yyyy)
	Monthly Pension Amount
	Annual Pension Amount

Spouse Pensioners

Required Data Fields	
To be provided for each year for FY 2022, FY 2023 & FY 2024 (Example- To be provided for the previous 3 years)	Beneficiary ID
	Name
	Category
	Gender
	Date of Birth (dd-mm-yyyy)
	Start Date (dd-mm-yyyy)
	Deceased Member Name
	Date Deceased (dd-mm-yyyy)
	Monthly Pension Amount
	Annual Pension Amount

Child Pensioners

Required Data Fields	
To be provided for each year for FY 2022, FY 2023 & FY 2024 (Example- To be provided for the previous 3 years)	Beneficiary ID
	Name
	Category
	Gender
	Date of Birth (dd-mm-yyyy)
	Start Date (dd-mm-yyyy)
	End Date (dd-mm-yyyy)
	Deceased Member Name
	Date Deceased (dd-mm-yyyy)
	Monthly Pension Amount
	Annual Pension Amount

Withdrawals Data

Required Data Fields	
	Employee ID
	Reason for Withdrawal (Retirement/Death/Deferred)

To be provided for each year for FY 2022, FY 2023 & FY 2024 (Example- To be provided for the previous 3 years)	Gender
	Date of Birth (dd-mm-yyyy)
	Date of Employment (dd-mm-yyyy)
	Date of Joining Scheme (dd-mm-yyyy)
	Date of Exit (dd-mm-yyyy)
	Monthly Pensionable Salary (as at scheme closure date)
	Monthly Pensionable Salary (at exit date)

Outstanding Benefits Payment

Required Data Fields	
To be provided for each year for FY 2022, FY 2023 & FY 2024 (Example- To be provided for the previous 3 years)	Member ID
	Member Category
	Gender
	Date of Birth (dd-mm-yyyy)
	Monthly Pension Amount
	Annual Pension Amount

Membership Register

Membership Register as at:

Year	
Scheme	Defined Benefit Scheme

Membership No	Member Name	Gender	Membership Status *(Active, Deferred, Retired, Dormant, Dead etc)*	Date of Birth	Date of Employment	Date of Joining Scheme	Final Average salary	Accumulated Member balances as at Scheme Closure	Accumulated Member balances as at December 2021	Accumulated Member balances as at December 2022	Accumulated Member balances as at December 2023	Accumulated Member balances as at December 2024

Deferred Member Data

Deferred Register as at:

Year	
Scheme	Defined Benefit Scheme

Member No	Member Name	Gender	Date of Birth	Date of Employment	Date of Joining Scheme	Date of Deferring the Scheme	Annual Salary at Date of Deferral	Deferred Amount (Lumpsum)	Deferred Amount (Pension)

Withdrawals Register

Withdrawals Register as at:

Year	
Scheme	Defined Benefit Scheme

Membership No	Member Name	Gender	Membership Status *(Deferred, Retired, Dormant, Dead etc)*	Date of Birth	Date of Employment	Date of Joining Scheme	Date of Exiting Scheme *(Not Applicable for Active Members)*	Reason for Exiting Scheme *(Death, Early/Normal/Ill Health Retirement, Leaving Employer etc)*	Amount Withdrawn *(if any)*	Date of Withdrawal *(if applicable)*

5. DATA CHECKS

5.1. Upon receiving data from Trustees for Defined Benefit (DB) and Defined Contribution (DC) valuation, it is important to conduct thorough data checks to ensure the accuracy and reliability of the information. Here are some essential data checks you should consider:

Reasonability & Appropriateness Checks

5.2. The following reasonability checks should be conducted on the data

a) Ensure all member data is complete, including personal information, employment history, and salary details.

b) Verify the accuracy of dates of birth, retirement, and any other relevant events.

c) Cross-reference beneficiary information and ensure it aligns with plan documents.

d) Verify the currency and units (e.g., annual, monthly) of reported salaries.

e) Validate service years and employment periods against historical records.

f) Verify the accuracy of contributions made by both employees and employers.

g) Ensure that contribution limits are not exceeded.

h) Confirm that vesting schedules are applied correctly.

i) Check for any unusual or incorrect withdrawals.

j) Member enrollment date is after Scheme Commencement date.

k) The information provided (membership, assets, investment return & audited financial statements) is relevant and represents the Scheme position as at Valuation Date.

l) Tests on reasonableness of pensionable salary, annual pension & membership (entry, exit & pensionable service).

Consistency Checks

5.3. The following consistency checks should be applied:

a) Check for consistency in salary growth rates and historical salary data.

b) Validate the use of appropriate discount rates, inflation rates, and other assumptions in actuarial calculations.

c) Check for consistency in benefit calculations, including factors such as final average salary, accrual rates, and indexing.

Practical Example

5.4. A practical example of how to conduct data checks for active and pensioner data can be accessed on this file: Data_Checks_Practical Example.xlsx

6. APPENDICES

Appendix 1: Sign-Off Template

Sign-off

Statement by the Lead Analyst

I confirm that I have carried out this work in line with the applicable KAFS Internal Technical Policy and Procedure. Below is a summary of the key assumptions made:

Key Assumptions

1 _____
2 _____
3 _____
4 _____
5 _____

Name: _____

Date: _____

Statement by Peer Reviewer

I have reviewed this work in line with the KAFS Work Review Policy and discussed all issues encountered with the Lead Analyst. Below are my review comments:

Review Comments

1 _____
2 _____
3 _____
4 _____
5 _____

Name: _____

Date: _____

Statement by Supervisor

I have reviewed this work and discussed the issues encountered and the key assumptions made by the Lead Analyst. I confirm that the results are in order.

Name: _____

Date: _____

7. APPROVAL

Thus, adopted and signed by the management and Chief Actuary of KAFS in January 2026.

APPROVED BY

Ezekiel Macharia

Chief Actuary & Managing Director – Kenbright Actuarial and Financial Services Limited

Date: January 2026

Signed:

kenbright

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